

Spring 2025

a)

$\Pi_{\text{hero-name}} (\sigma_{\text{homeworld} = \text{'Earth'}} (\text{Hero}))$

b)

$\Pi_{\text{hero-name}} \left(\rho_H (\text{Hero}) \bowtie_{H.\text{hero-id} = A.\text{hero-id}} \rho_A (\text{Assignment}) \right)$

OR

$\Pi_{H.\text{hero-name}} \left(\sigma_{T.\text{team-name} = \text{'Avengers'}} \left(\rho_H (\text{Hero}) \bowtie_{H.\text{hero-id} = T.\text{leader-id}} \rho_T (\text{Team}) \right) \right)$

c) $\Pi_{H.\text{hero-name}} ($

$\sigma_{T.\text{headquarters} = \text{'Wakanda'}} \left(\rho_H (\text{Hero}) \bowtie_{H.\text{hero-id} = T.\text{leader-id}} \rho_T (\text{Team}) \right)$

Fall 2024

a)

$\left(\begin{array}{l} \sigma_{\text{gender} = 'male'} (\text{Patients}) \\ \wedge \text{address} = 'Palashi' \end{array} \right)$

b)

$\pi_{P.\text{name}, M.\text{name}} \left(\begin{array}{l} \rho_P (\text{Patients}) \bowtie P.\text{patient_id} = A.\text{patient_id} \rho_A (\text{Appointments}) \\ \bowtie A.\text{appointment_id} = P.\text{appointment_id} \rho_P (\text{Prescriptions}) \\ \bowtie P.\text{prescriptions_id} = M.\text{prescription_id} \rho_M (\text{Medications}) \\ \bowtie M.\text{medicine_id} = M.\text{medicine_id} \rho_M (\text{Medicine}) \end{array} \right)$

c)

$\pi_{D.\text{specialization}}$

$\sigma_{P.\text{patient_id} = \text{NULL}} \left(\rho_D (\text{Doctors}) \right)$

$\bowtie D.\text{doctor_id} = A.\text{doctor_id} \rho_A (\text{Appointments})$

$\bowtie A.\text{patient_id} = P.\text{patient_id} \rho_P (\text{Patients})$

$\wedge P.\text{age} > 70 \quad))$

Course Code

Trimester / Semester : Spring / Summer / Fall, 20.....

Name of Exam : Class Test / Mid-term / Final

Date:

OR
Storing output in $\mathcal{G}_x$
 $\mathcal{G}_x = \Pi A.doctor-id$
 $\mathcal{G}_x \rightarrow \mathcal{G}_x$

($\sigma_{P.age > 70}$ ($P_A(Appointments)$))
 $\mathcal{A}.patient-id = P.patient-id$ ($P(Patients)$)

$\Pi D.specialization(\sigma_{D.doctor-id \neq \mathcal{G}_x} (P(Doctors)))$